

Time to think

Kuo was brilliant at solving problems, but things didn't always go well. After working as a military leader, Kuo was blamed when 60,000 soldiers were killed by enemy forces. He was banished from the center of government to a house he had built in the countryside.

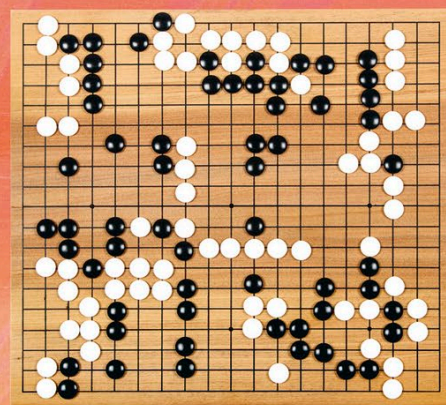
For a deep thinker, this was bliss. Kuo happily began writing down all his scientific ideas about the world, creating a book called *Mengxi bitan* (Brush Talks from Dream Pool). It includes essays on dozens of different areas of nature—from the distant stars, to the rocks, animals, and plants he saw on his travels.

Many of Kuo's ideas were hundreds of years ahead of the rest of the world. For instance, he wrote about his worry that the environment was damaged when forests were cleared for wood to burn. Kuo's book is also the first to mention that the Earth's climate changes over time.

Board-game genius

Math-whiz Kuo figured out how many different combinations of counters are possible on an ancient board game called Go. He arrived at 10^{172} , which is 10 with 171 zeros after it!

No one had ever calculated such a huge number before!



He found fossilized bamboo in an area where the weather wasn't suitable for bamboo growth. Kuo realized the climate had changed over time.